

Introducción a Tomcat 10

1. Instalación de Tomcat 10

sudo apt update

sudo apt install tomcat10

```
Pestanya nova  Dividix la vista  Copia  Apega  Busca  ≡
serralta@serralta:~$ sudo apt install tomcat10
[sudo] password for serralta:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java default-jre-headless java-common libapr1t64
  libasound2-data libasound2t64 libavahi-client3 libavahi-common-data libavahi-common3 libcups2t64
  libecjclipse-jdt-core-java libgraphite2-3 libharfbuzz0b liblcms2-2 libpcsclite1 libtclnative-1
  libtomcat10-java openjdk-21-jre-headless tomcat10-common
Suggested packages:
  default-jre alsa-utils libasound2-plugins cups-common liblcms2-utils pscd libnss-mdns
  fonts-dejavu-extra fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhei | fonts-wqy-zenhei
  fonts-indic tomcat10-admin tomcat10-docs tomcat10-examples tomcat10-user
The following NEW packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java default-jre-headless java-common libapr1t64
  libasound2-data libasound2t64 libavahi-client3 libavahi-common-data libavahi-common3 libcups2t64
  libecjclipse-jdt-core-java libgraphite2-3 libharfbuzz0b liblcms2-2 libpcsclite1 libtclnative-1
  libtomcat10-java openjdk-21-jre-headless tomcat10 tomcat10-common
0 upgraded, 22 newly installed, 0 to remove and 104 not upgraded.
Need to get 61.0 MB of archives.
After this operation, 225 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://es.archive.ubuntu.com/ubuntu/noble/main amd64 alsa-topology-conf all 1.2.5.1-2 [15.5 kB]
Get:2 http://es.archive.ubuntu.com/ubuntu/noble-updates/main amd64 libasound2-data all 1.2.11-1ubuntu0.
1 [21.1 kB]
Get:3 http://es.archive.ubuntu.com/ubuntu/noble-updates/main amd64 libasound2t64 amd64 1.2.11-1ubuntu0.
1 [399 kB]
```

2. Gestión del servicio Tomcat

sudo systemctl status tomcat10

sudo systemctl stop tomcat10

sudo systemctl start tomcat10

sudo systemctl restart tomcat10

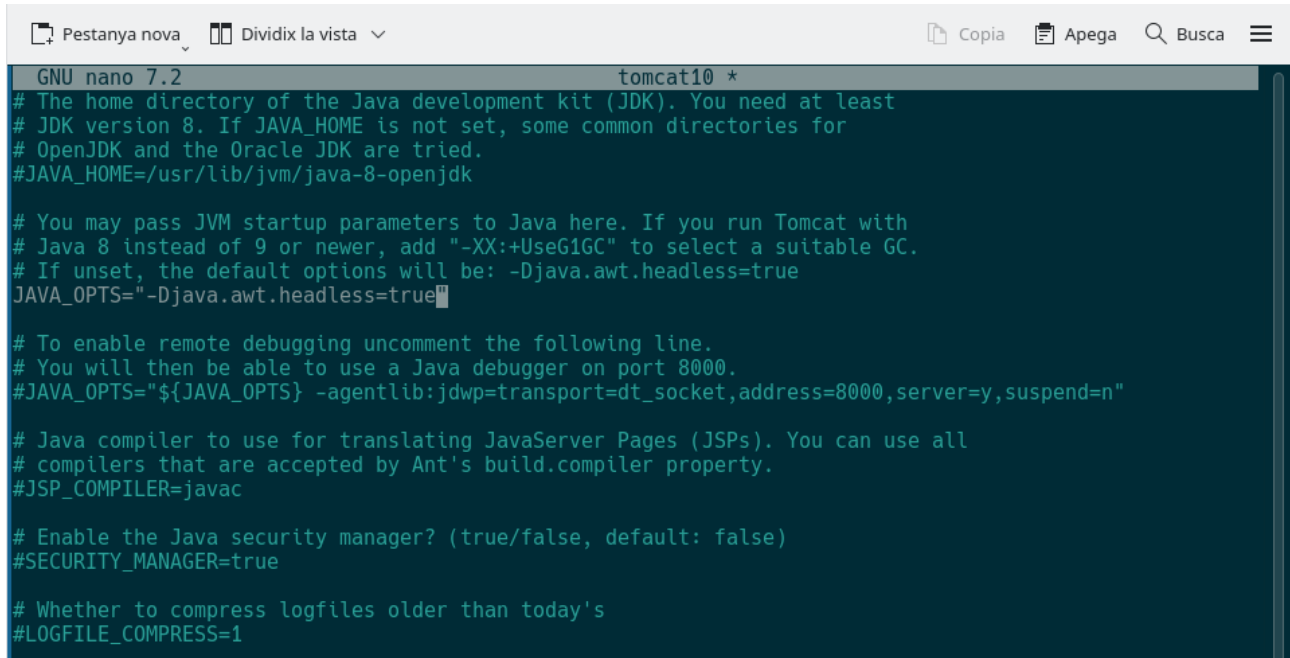
```
serralta@serralta:~$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Server
   Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-11-18 12:18:25 UTC; 3min 48s ago
     Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
   Process: 2939 ExecStartPre=/usr/libexec/tomcat10/tomcat-update-policy.sh (code=exited, status=0/SUCCESS)
   Main PID: 2944 (java)
     Tasks: 29 (limit: 4606)
    Memory: 84.3M (peak: 84.5M)
       CPU: 5.800s
   CGroup: /system.slice/tomcat10.service
           └─2944 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.file=/var/lib/tomcat10/conf/logging.properties

Nov 18 12:18:27 serralta tomcat10[2944]: OpenSSL inicializado correctamente [OpenSSL 3.0.13 30 Jan 2025]
Nov 18 12:18:29 serralta tomcat10[2944]: Inicializando el manejador de protocolo ["http-nio-8080"]
Nov 18 12:18:29 serralta tomcat10[2944]: Server initialization in [3367] milliseconds
Nov 18 12:18:30 serralta tomcat10[2944]: Arrancando servicio [Catalina]
Nov 18 12:18:30 serralta tomcat10[2944]: Starting Servlet engine: [Apache Tomcat/10.1.16 (Ubuntu)]
Nov 18 12:18:30 serralta tomcat10[2944]: Desplegando el directorio [/var/lib/tomcat10/webapps/ROOT] de
Nov 18 12:18:32 serralta tomcat10[2944]: Al menos un JAR, que se ha explorado buscando TLDs, aún no co
Nov 18 12:18:32 serralta tomcat10[2944]: Deployment of web application directory [/var/lib/tomcat10/we
Nov 18 12:18:32 serralta tomcat10[2944]: Starting ProtocolHandler ["http-nio-8080"]
Nov 18 12:18:32 serralta tomcat10[2944]: Server startup in [3001] milliseconds
lines 1-22/22 (END)
```

3. Opciones del servicio (JAVA_OPTS)

sudo nano /etc/default/tomcat10

JAVA_OPTS="-Djava.awt.headless=true"



```
GNU nano 7.2 tomcat10 *
# The home directory of the Java development kit (JDK). You need at least
# JDK version 8. If JAVA_HOME is not set, some common directories for
# OpenJDK and the Oracle JDK are tried.
#JAVA_HOME=/usr/lib/jvm/java-8-openjdk

# You may pass JVM startup parameters to Java here. If you run Tomcat with
# Java 8 instead of 9 or newer, add "-XX:+UseG1GC" to select a suitable GC.
# If unset, the default options will be: -Djava.awt.headless=true
JAVA_OPTS="-Djava.awt.headless=true"

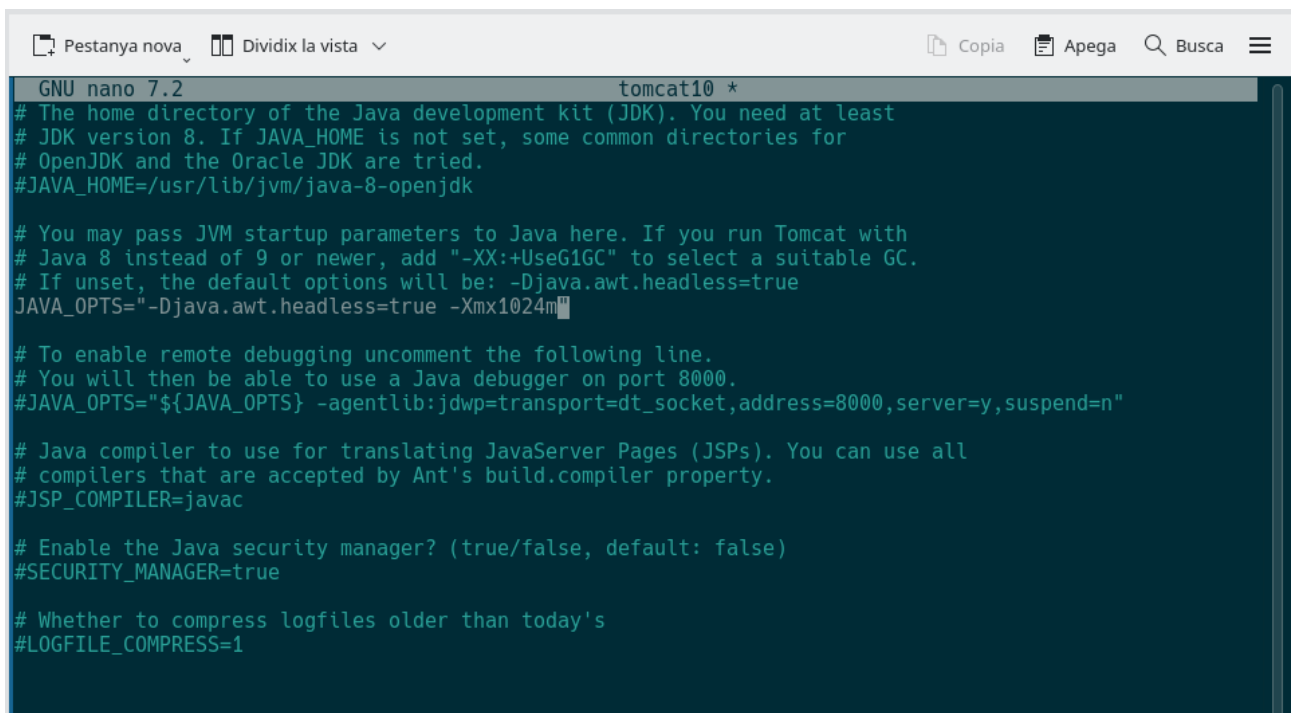
# To enable remote debugging uncomment the following line.
# You will then be able to use a Java debugger on port 8000.
#JAVA_OPTS="${JAVA_OPTS} -agentlib:jdwp=transport=dt_socket,address=8000,server=y,suspend=n"

# Java compiler to use for translating JavaServer Pages (JSPs). You can use all
# compilers that are accepted by Ant's build.compiler property.
#JSP_COMPILER=javac

# Enable the Java security manager? (true/false, default: false)
#SECURITY_MANAGER=true

# Whether to compress logfiles older than today's
#LOGFILE_COMPRESS=1
```

JAVA_OPTS="-Djava.awt.headless=true -Xmx1024m"



```
GNU nano 7.2 tomcat10 *
# The home directory of the Java development kit (JDK). You need at least
# JDK version 8. If JAVA_HOME is not set, some common directories for
# OpenJDK and the Oracle JDK are tried.
#JAVA_HOME=/usr/lib/jvm/java-8-openjdk

# You may pass JVM startup parameters to Java here. If you run Tomcat with
# Java 8 instead of 9 or newer, add "-XX:+UseG1GC" to select a suitable GC.
# If unset, the default options will be: -Djava.awt.headless=true
JAVA_OPTS="-Djava.awt.headless=true -Xmx1024m"

# To enable remote debugging uncomment the following line.
# You will then be able to use a Java debugger on port 8000.
#JAVA_OPTS="${JAVA_OPTS} -agentlib:jdwp=transport=dt_socket,address=8000,server=y,suspend=n"

# Java compiler to use for translating JavaServer Pages (JSPs). You can use all
# compilers that are accepted by Ant's build.compiler property.
#JSP_COMPILER=javac

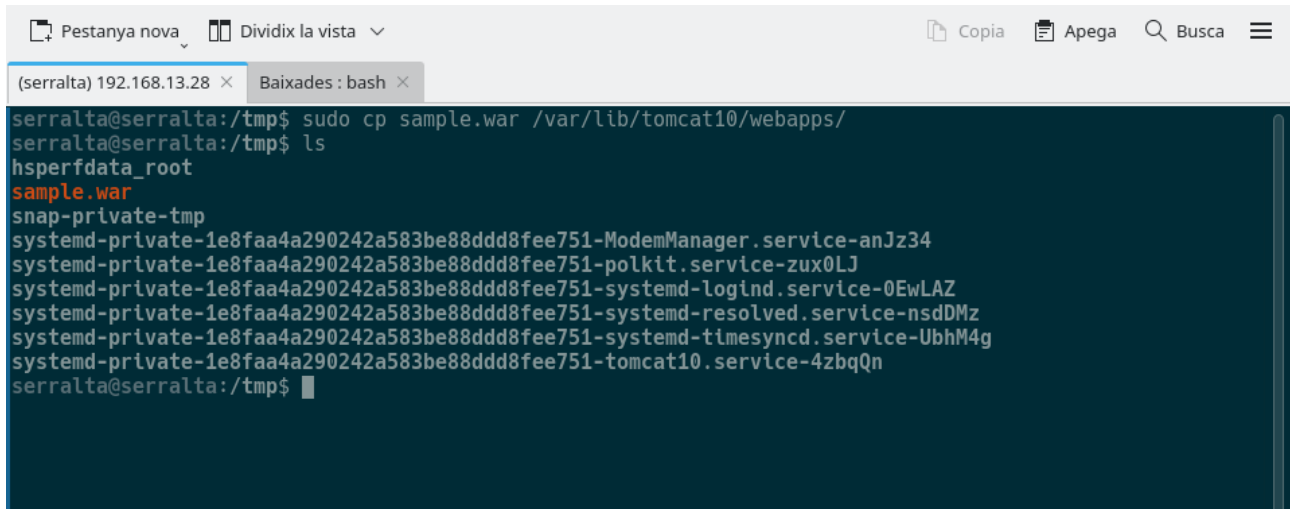
# Enable the Java security manager? (true/false, default: false)
#SECURITY_MANAGER=true

# Whether to compress logfiles older than today's
#LOGFILE_COMPRESS=1
```

sudo systemctl restart tomcat10

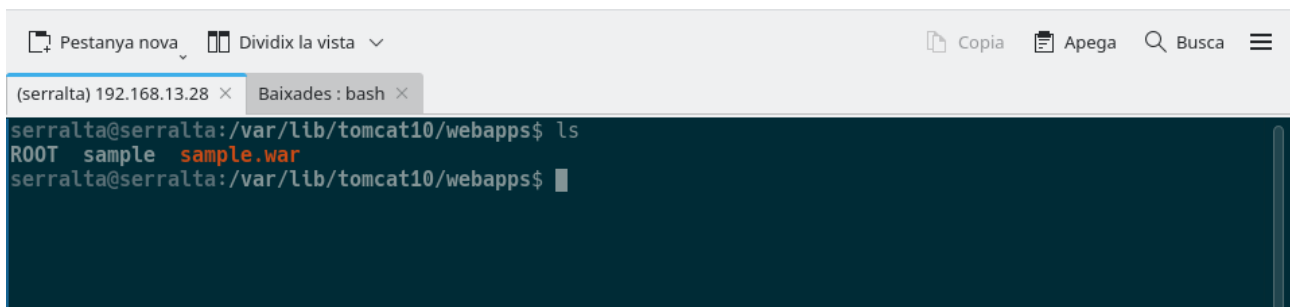
4. Despliegue de aplicaciones desde la terminal

`sudo cp sample.war /var/lib/tomcat10/webapps/`



A terminal window titled "Baixades : bash" showing the execution of the command `sudo cp sample.war /var/lib/tomcat10/webapps/`. The prompt is `serralta@serralta:/tmp$`. The command is executed, and the prompt changes to `serralta@serralta:/tmp$`. The user then runs `ls`, and the output lists the contents of the current directory, including `sample.war` in red text.

```
serralta@serralta:/tmp$ sudo cp sample.war /var/lib/tomcat10/webapps/
serralta@serralta:/tmp$ ls
hsperfdata_root
sample.war
snap-private-tmp
systemd-private-1e8faa4a290242a583be88ddd8fee751-ModemManager.service-anJz34
systemd-private-1e8faa4a290242a583be88ddd8fee751-polkit.service-zux0LJ
systemd-private-1e8faa4a290242a583be88ddd8fee751-systemd-logind.service-0EwLAZ
systemd-private-1e8faa4a290242a583be88ddd8fee751-systemd-resolved.service-nsdDMz
systemd-private-1e8faa4a290242a583be88ddd8fee751-systemd-timesyncd.service-UbhM4g
systemd-private-1e8faa4a290242a583be88ddd8fee751-tomcat10.service-4zbqQn
serralta@serralta:/tmp$
```

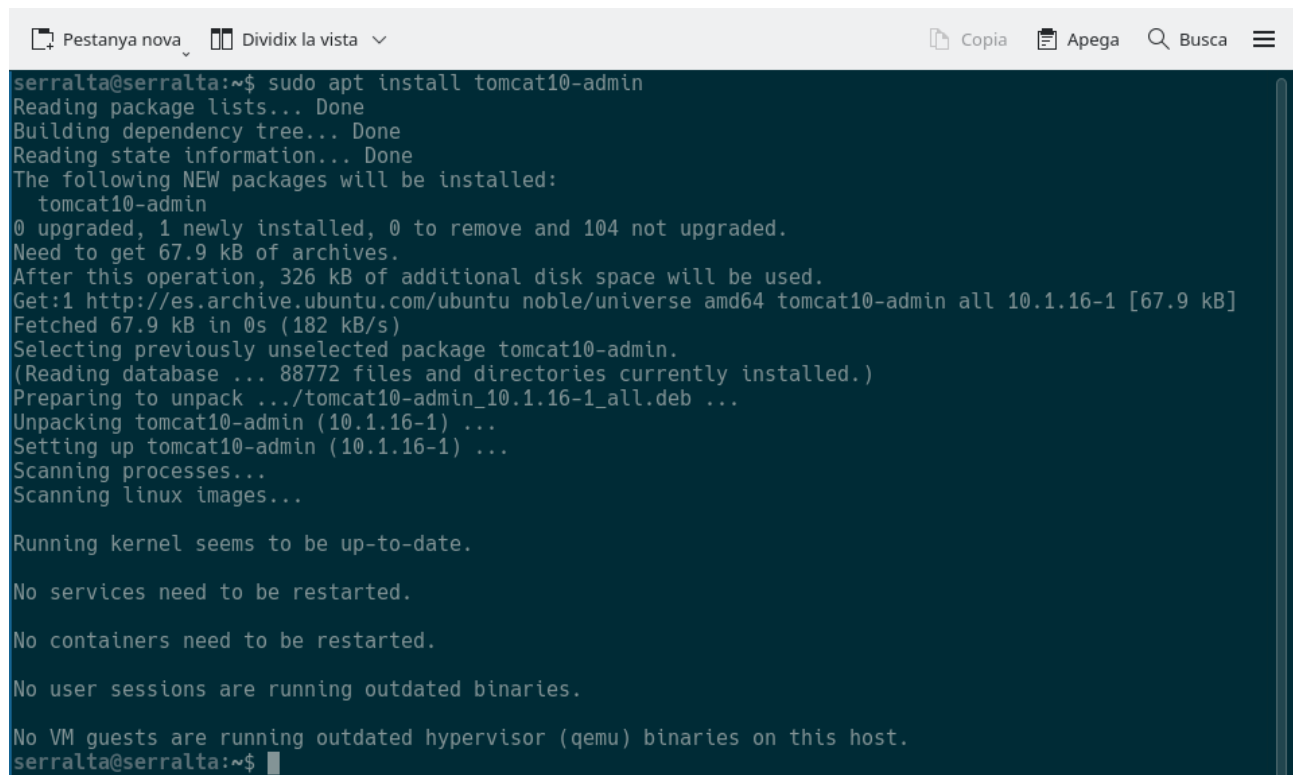


A terminal window titled "Baixades : bash" showing the verification of the deployment. The prompt is `serralta@serralta:/var/lib/tomcat10/webapps$`. The user runs `ls`, and the output shows `sample.war` in red text.

```
serralta@serralta:/var/lib/tomcat10/webapps$ ls
ROOT sample sample.war
serralta@serralta:/var/lib/tomcat10/webapps$
```

5. Instalación de Tomcat Manager (interfaz web)

sudo apt install tomcat10-admin

A terminal window with a dark blue background and white text. The window has a title bar with 'Pestanya nova' and 'Dividix la vista'. The terminal output shows the command 'sudo apt install tomcat10-admin' being executed. It lists the packages to be installed, the disk space requirements, and the progress of the installation. The installation is successful, and the terminal shows the prompt 'serralta@serralta:~\$' at the end.

```
serralta@serralta:~$ sudo apt install tomcat10-admin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  tomcat10-admin
0 upgraded, 1 newly installed, 0 to remove and 104 not upgraded.
Need to get 67.9 kB of archives.
After this operation, 326 kB of additional disk space will be used.
Get:1 http://es.archive.ubuntu.com/ubuntu/noble/universe amd64 tomcat10-admin all 10.1.16-1 [67.9 kB]
Fetched 67.9 kB in 0s (182 kB/s)
Selecting previously unselected package tomcat10-admin.
(Reading database ... 88772 files and directories currently installed.)
Preparing to unpack .../tomcat10-admin_10.1.16-1_all.deb ...
Unpacking tomcat10-admin (10.1.16-1) ...
Setting up tomcat10-admin (10.1.16-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

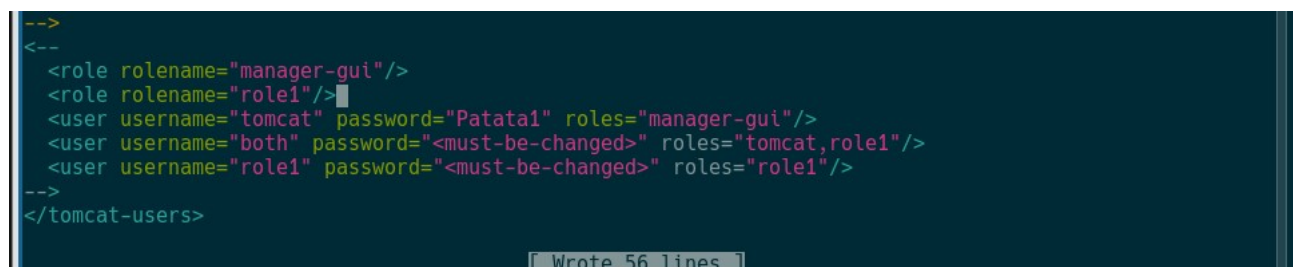
No VM guests are running outdated hypervisor (qemu) binaries on this host.
serralta@serralta:~$
```

6-Crear usuario administrador

sudo nano /etc/tomcat10/tomcat-users.xml

<role rolename="manager-gui"/>

<user username="tomcat" password="s3cret" roles="manager-gui"/>

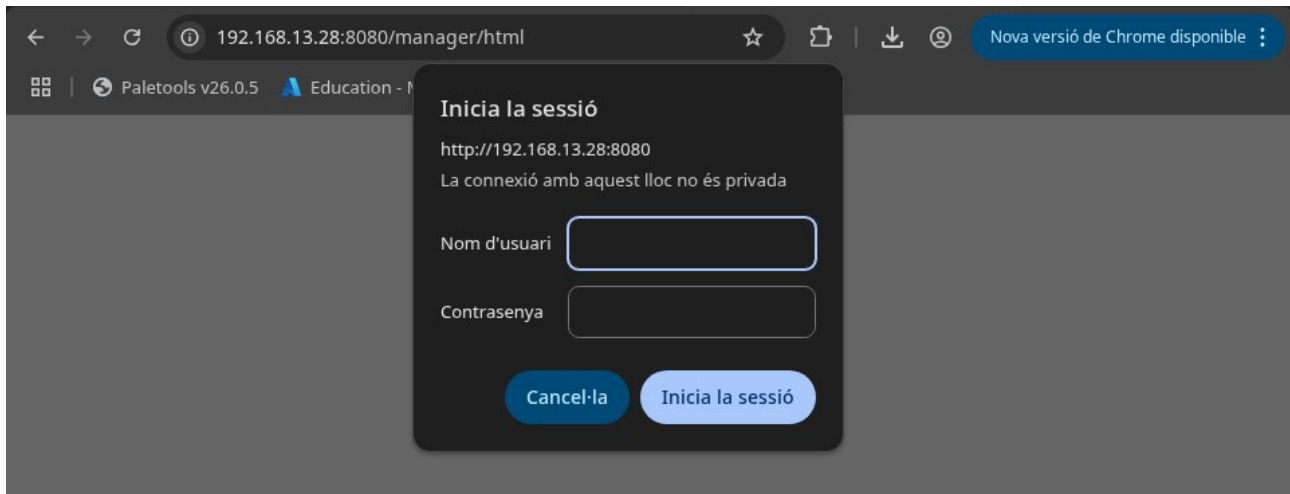
A terminal window with a dark blue background and white text. The window shows the contents of the file /etc/tomcat10/tomcat-users.xml. The text is color-coded: XML tags are in green, attributes are in yellow, and values are in white. The file content includes a role named 'manager-gui' and a user named 'tomcat' with the password 'Patata1' and roles 'manager-gui' and 'role1'. The terminal shows the prompt 'serralta@serralta:~\$' at the end.

```
-->
<--
<role rolename="manager-gui"/>
<role rolename="role1"/>
<user username="tomcat" password="Patata1" roles="manager-gui"/>
<user username="both" password="<must-be-changed>" roles="tomcat,role1"/>
<user username="role1" password="<must-be-changed>" roles="role1"/>
-->
</tomcat-users>

[ Wrote 56 lines ]
```

sudo systemctl restart tomcat10

7. Acceso a la interfaz web de administración



The screenshot shows the Tomcat Manager web interface. The browser address bar displays `192.168.13.28:8080/manager/html`. The page features the Tomcat logo on the left and the Apache Software Foundation logo on the right. The main heading is "Gestor de Aplicaciones Web de Tomcat". Below the heading, there is a message box showing "Mensaje: OK". The interface is divided into two main sections: "Gestor" and "Aplicaciones".

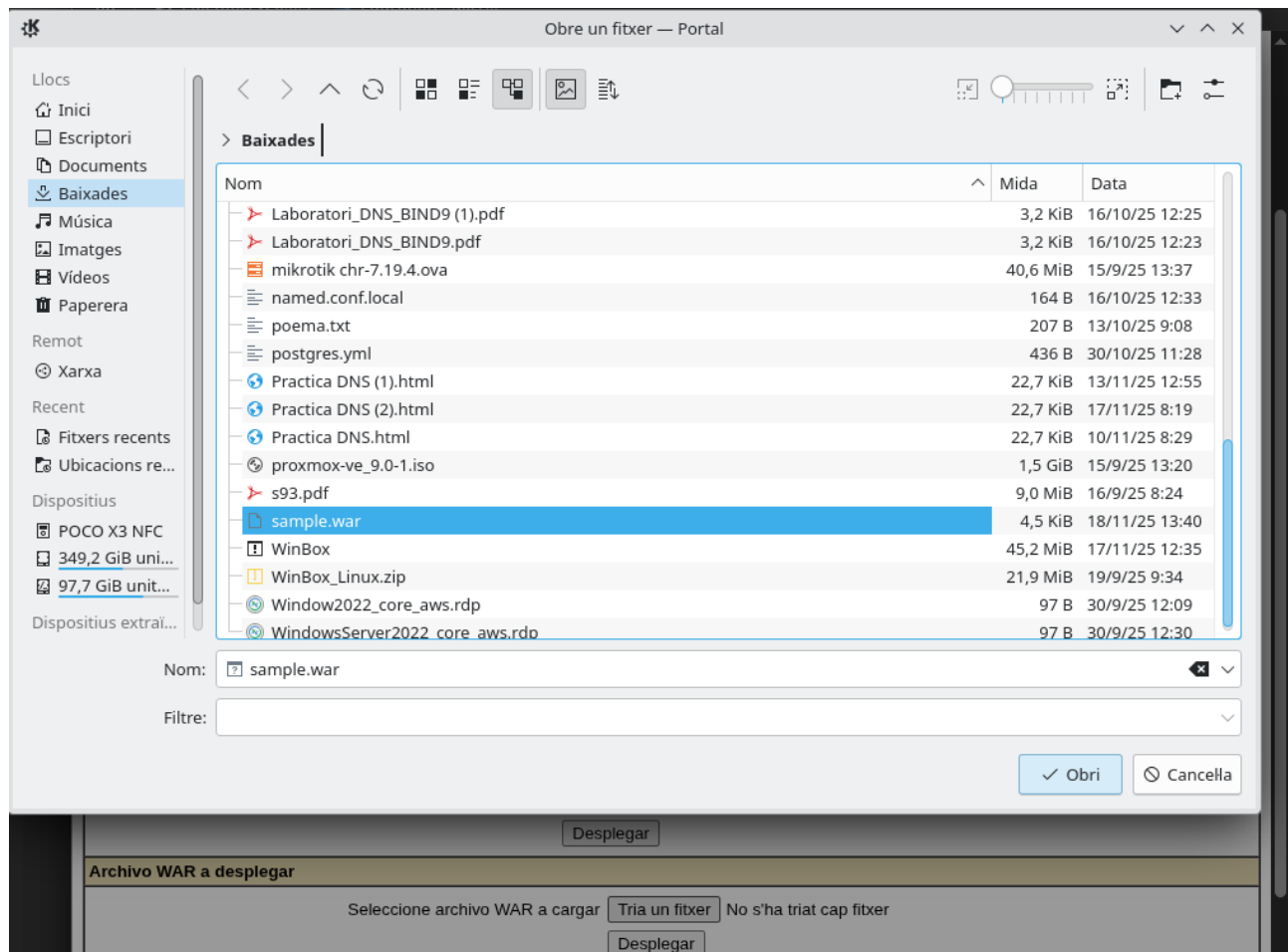
Gestor

Listar Aplicaciones	Ayuda HTML de Gestor	Ayuda de Gestor	Estado de Servidor
-------------------------------------	--------------------------------------	---------------------------------	------------------------------------

Aplicaciones

Ruta	Versión	Nombre a Mostrar	Ejecutándose	Sesiones	Comandos
/	Ninguno especificado		true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/host-manager	Ninguno especificado	Tomcat Host Manager Application	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos

8. Despliegue de aplicaciones mediante interfaz web



Desplegar

Desplegar directorio o archivo WAR localizado en servidor

Trayectoria de Contexto (opcional):

Version (for parallel deployment):

URL de archivo de Configuración XML:

URL de WAR o Directorio:

Desplegar

Archivo WAR a desplegar

Seleccione archivo WAR a cargar Tria un fitxer sample.war

Desplegar

Mensaje:

OK

9. Archivos de configuración importantes

cat /etc/tomcat10/server.xml

```
serralta@serralta:~$ sudo cat /etc/tomcat10/server.xml
<?xml version="1.0" encoding="UTF-8"?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<!-- Note: A "Server" is not itself a "Container", so you may not
define subcomponents such as "Valves" at this level.
Documentation at /docs/config/server.html
-->
<Server port="-1" shutdown="SHUTDOWN">
  <Listener className="org.apache.catalina.startup.VersionLoggerListener" />
  <!-- Security listener. Documentation at /docs/config/listeners.html
  <Listener className="org.apache.catalina.security.SecurityListener" />
  -->
  <!-- APR library loader. Documentation at /docs/apr.html -->
  <Listener className="org.apache.catalina.core.AprLifecycleListener" SSLEngine="on" />
  <!-- Prevent memory leaks due to use of particular java/javax APIs-->
  <Listener className="org.apache.catalina.core.JreMemoryLeakPreventionListener" />
  <Listener className="org.apache.catalina.mbeans.GlobalResourcesLifecycleListener" />
  <Listener className="org.apache.catalina.core.ThreadLocalLeakPreventionListener" />

  <!-- Global JNDI resources
  Documentation at /docs/jndi-resources-howto.html
  -->
  <GlobalNamingResources>
    <!-- Editable user database that can also be used by
    UserDatabaseRealm to authenticate users
    -->
    <Resource name="UserDatabase" auth="Container"
      type="org.apache.catalina.UserDatabase"
      description="User database that can be updated and saved"
      factory="org.apache.catalina.users.MemoryUserDatabaseFactory"
      pathname="conf/tomcat-users.xml" />
  </GlobalNamingResources>
```

cat /etc/tomcat10/context.xml

```
Pestanya nova  Dividix la vista  Copia  Apega  Busca  ≡
(serralta) 192.168.13.28  Baixades: bash  X
serralta@serralta:~$ sudo cat /etc/tomcat10/context.xml
<?xml version="1.0" encoding="UTF-8"?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<!-- The contents of this file will be loaded for each web application -->
<Context>

    <!-- Default set of monitored resources. If one of these changes, the    -->
    <!-- web application will be reloaded.                                -->
    <WatchedResource>WEB-INF/web.xml</WatchedResource>
    <WatchedResource>WEB-INF/tomcat-web.xml</WatchedResource>
    <WatchedResource>${catalina.base}/conf/web.xml</WatchedResource>

    <!-- Uncomment this to enable session persistence across Tomcat restarts -->
    <!--
    <Manager pathname="SESSIONS.ser" />
    -->
</Context>
serralta@serralta:~$
```

cat /etc/tomcat10/web.xml

```
<!-- ===== Default Welcome File List ===== -->
<!-- When a request URI refers to a directory, the default servlet looks -->
<!-- for a "welcome file" within that directory and, if present, to the -->
<!-- corresponding resource URI for display. -->
<!-- If no welcome files are present, the default servlet either serves a -->
<!-- directory listing (see default servlet configuration on how to -->
<!-- customize) or returns a 404 status, depending on the value of the -->
<!-- listings setting. -->
<!--
<!-- If you define welcome files in your own application's web.xml -->
<!-- deployment descriptor, that list *replaces* the list configured -->
<!-- here, so be sure to include any of the default values that you wish -->
<!-- to use within your application. -->

    <welcome-file-list>
        <welcome-file>index.html</welcome-file>
        <welcome-file>index.htm</welcome-file>
        <welcome-file>index.jsp</welcome-file>
    </welcome-file-list>

</web-app>
serralta@serralta:~$
```